

Personal 3D Printing without Visual Dependency in the Generative AI Era: Opportunities and Challenges

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Abstract

The 3D design and desktop 3D printing personal fabrication ecosystems are largely structured around visual interaction, limiting meaningful access for Blind and Low-Vision (BLV) users. This work presents first-person observations in which BLV researchers engaged in an exploratory investigation into an AI-mediated personal fabrication workflow with text-guided 3D generation, conversational inspection, and artifact printing. All of these system features were constructed with general-purpose AI capabilities to reflect an approach that builds on widely available AI features to support accessibility needs. Through the implementation and first-person experience of this AI-integrated workflow, we identify limitations in dimension control, reference-based editing, and post-print interpretation when relying on such AI-driven features. We reflect on these limitations to surface challenges in relying on general-purpose AI functionality while outlining preliminary design implications that provide a formative grounding for future research. This work envisions more ubiquitous and independent personal fabrication practices for BLV users in the era of generative AI.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; *Accessibility design and evaluation methods*.

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Keywords

Assistive Technology, Personal Fabrication, Blind and Low Vision, Research through Design, Generative AI, Conversational Interface

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1 Introduction

The rise of 3D printing technologies has brought personal fabrication to a desktop level for everyday design and engineering prototyping activities. However, its typical workflows, which span model design, inspection, slicing, and printer operation, remain heavily dependent on computer-based visualization, creating a fundamental barrier for Blind and Low-Vision (BLV) users. Recent advances in Vision–Language Model (VLM)–supported chatbots and generative 3D modeling offer opportunities to reduce this dependency by enabling semantically driven 3D creation and conversational access to visual information. When integrated into existing 3D printing ecosystems, these general-purpose Artificial Intelligence (AI) capabilities have the potential to provide an accessible pathway toward independent personal fabrication for BLV users.

Therefore, it is important to examine how such AI features function within personal fabrication workflows and where limitations emerge. Contributions presented in this work are derived from a preliminary exploration and formative reflection process in evaluating example design experiences with AI support. This includes two interconnected layers of AI-supported 3D design interactions through chatbots and generative models, as well as the broader AI-mediated end-to-end fabrication workflow that spans design, fabrication, and physical making. We intentionally adopt standard, general-purpose AI features to examine how widely available AI



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capabilities can provide timely and low-threshold support for accessible personal fabrication. We expect these findings to provide actionable knowledge that can inform the design of accessible features and our subsequent prototype iterations in future studies, contributing to the extension of independent personal fabrication for people with diverse accessibility needs within the generative AI era.

In summary, we conducted our research following a *Research through Design* (RtD) [7] process and contributed in the following areas:

- An early-stage *Design* of AI-mediated accessible personal fabrication workflows for BLV users, situated in the use of general-purpose AI features.
- *Reflections* that surface preliminary design implications for addressing the observed limitations in such AI-mediated workflows.

2 Related Work

2.1 Personal Fabrication for BLV Users

The artifacts produced by 3D printing offer a wide range of use cases for BLV users. These include superimposing tactile patterns onto 3D models [10], extending 2D tactile graphic maps into 3D forms [5], prototyping electro-mechanical artifacts for an accessible context [9], and providing tangible geometric references for educational scenarios [4]. However, relatively fewer studies have examined how BLV users can better use 3D printing itself as a tool for independent personal fabrication. Recent efforts, such as *Guidelines for Producing Accessible 3D Prints (2024)* [3], offer initial guidance for supporting BLV users throughout the personal fabrication workflow, with attention to accessibility considerations at different stages. Other research has primarily focused on enabling printer operation through simplified interaction or integrated electronic components [1, 2], but has limited the design stage to selecting only pre-designed models. Our work emphasizes the process of independent design and evaluation, while situating it within complete personal fabrication workflows.

2.2 Non-visual 3D Design Interfaces

3D design is an iterative loop of action and inspection, and much related research has focused on how software and hardware systems can support this interaction loop for BLV users. *ShapeCAD* [11] employs a 2.5D display to render portions of a digital model for tactile exploration, and other studies have used commercial kinesthetic devices combined with audio feedback to convey the geometric features of 3D meshes [6]. Although these approaches excel at delivering multimodal real-time feedback, their reliance on specialized haptic hardware limits their applicability to ubiquitous personal fabrication settings.

A11yShape [14] presents an AI-assisted modeling environment (using script-based OpenSCAD) that enables BLV users to engage in the design loop through basic programming and conversational interaction as a low-cost, software-driven approach. Building on a similar conceptual foundation, our work shifts toward a lower-threshold expectation by emphasizing purely semantic, text-driven operation that minimizes technical skill dependency. This paradigm

is largely based on text-guided 3D diffusion processes [13] as the primary strategy for model generation.

3 Study Overview

3.1 Collaborative Setting

In this study, our team consisted of blind and sighted researchers working in a collaborative exploration setting. The sighted researchers focused on the implementation of the AI-mediated system, while the blind researchers guided the system's use scenarios, design intentions, and the selection of technological approaches based on their lived experiences.

This work focuses on enabling independent personal fabrication for BLV users using commercial 3D printers and personal electronic devices. The observations and suggestions reported here reflect the first-person evaluation of technical systems embedded in the RtD process, rather than data collected from external participants.

3.2 System Flow and AI Tooling

The complete system workflow of our study is demonstrated in **Figure 1**. We utilized the Hunyuan3D 2.5 model [12] to generate 3D designs, and the multi-modal capabilities of the GPT-5.2 model [8] to build a chatbot for inspecting clips of the 3D designs. Since there is no mature tool for direct 3D format file editing, we implemented a detour method by semantically editing the clips of the model with the GPT Image-1.5 model, then re-generating a 3D design with these edits (shown in the workflow diagram's top-right panel). Furthermore, we implemented all feature operations in an HTML front-end and enabled the text-to-speech function using the Web Speech API.

Alongside, we implemented a custom printer code generator to produce three canonical orthographic projections of given 3D models, as a parallel method to Lieb et al.'s work [6] rendered through haptic interfaces. This allows us to contrast how similar shape information can be leveraged on our alternative platforms by providing these projections as flat tactile prints for pre-print inspection. The rationale is that silhouette contours capture key geometric features and overall proportions while requiring significantly less time and material than printing a complete model. Such patterns are evaluated once in the *Physical Inspection* stage and are involved in one core discussion section.

4 Exploratory Design Process

We explored two design-fabrication flows using the designed system, where the artifacts involved in each flow are indicated by ① and ② in **Figure 1**, respectively.

We initiated each flow with semantic design intentions:

"I am developing a 3D shooting game designed for blind users. I would like you to help me design a mascot character for this game. [1①]"

"Help me design a keychain with a circular flat shape of 4 cm in diameter. Add a 0.9 cm circular hole slightly above the center for attaching a key ring. In the center, create a 0.6 cm thick embossed Maine Coon cat element that can be recognized through touch. Leave a 0.4 cm

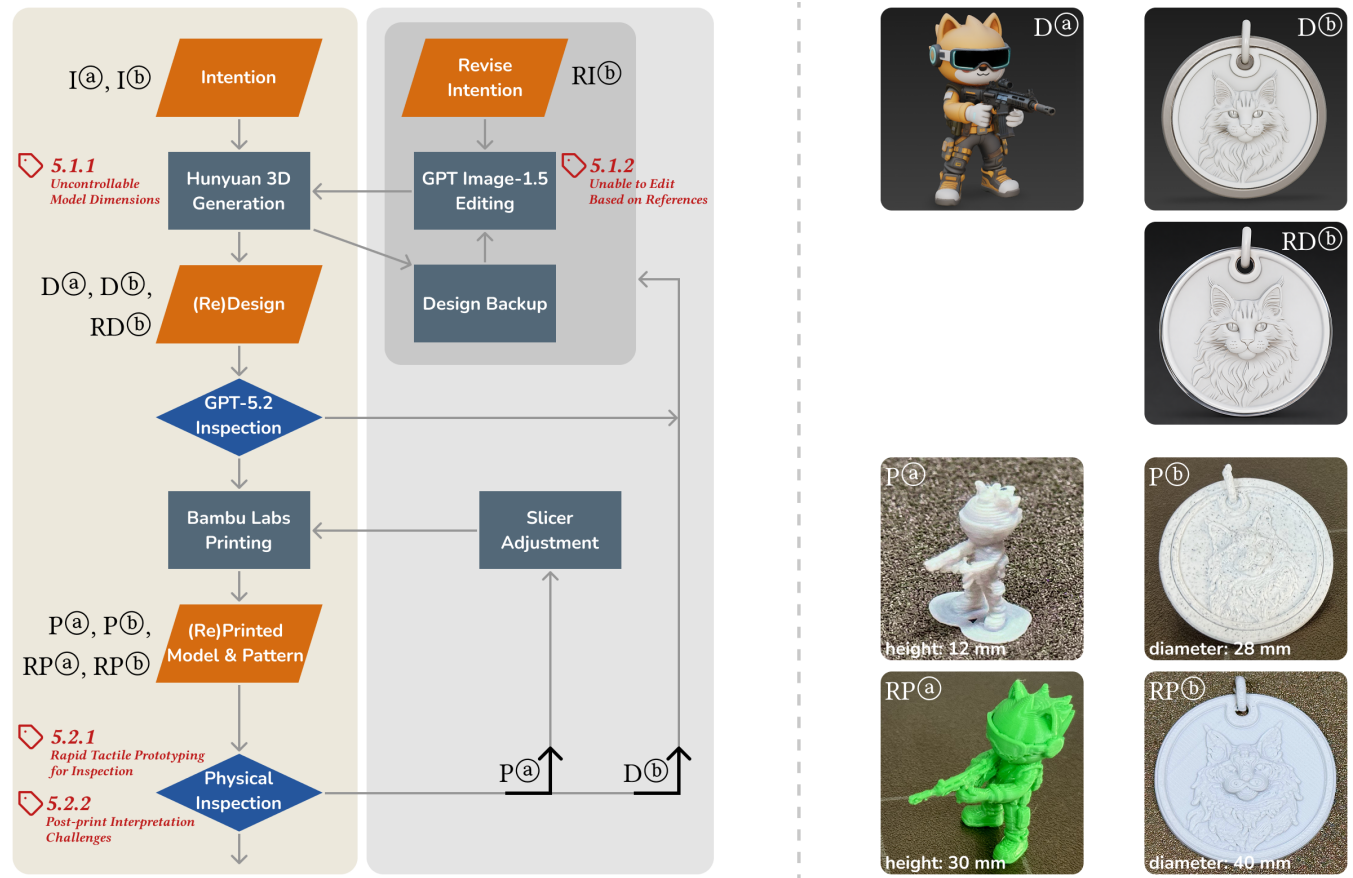


Figure 1: System technical stack and interaction workflow of our study. Flow Diagram: The left panel illustrates the primary design-fabrication pipeline, while the right panel presents the auxiliary editing pathway. Capital-circled annotations (e.g., I(a), D(b)) denote the discussed artifacts within the workflow, correlating to their adjacent processing blocks. Tag labels indicate the key discussion sections, each aligned with their adjacent block stages. Artifact Photos: The images of the discussed artifacts that have visual representations (digital and printed models).

margin around the edge and smooth the edge with a 0.2 cm curved surface. [I(b)]

After generating D(a) and D(b) from these intentions, we engaged in a conversational inspection process lasting approximately five minutes, during which we descriptively examined the generated designs to form an initial, pre-fabrication understanding. None of the flows chose to edit the model at this stage, although they were allowed.

The models were sliced with the Bambu Lab slicer before being wirelessly sent to the printer. The printing processes were performed on standard Bambu Lab A1 machines, which produced P(a) and P(b).

After physical inspections, we modified the two designs separately. D(a)'s change was only a simple slicer adjustment, while D(b) underwent the editing flow with the intention below:

"Remove the outer circular rim and redesign the edge inspired by the curved edge of Huawei waterfall display screens. Do not modify other parts." [RI(b)] (We tried to apply a fillet to the top edge)

The modification process returned to the main workflow and produced RD(b), RP(a), and RP(b), similar to the process described above. RP(a) and RP(b) were also processed using our silhouette tactile pattern generator for the aforementioned inspection comparison.

5 Design Reflections and Implications

The flow diagram in **Figure 1** also contains labels for the majority of discussion sections along with their positioned workflow blocks. Each section concludes with a **Reflection** that surfaces preliminary design implications for each situation.

5.1 Generative Modeling Limitations

Overall, the AI-assisted workflow enabled the rapid initiation of personal designs, particularly for non-parametric and everyday design tasks. Both generated designs aligned with our initial design intentions during conversational inspection. We observed some issues in implementing the generative 3D models, as discussed in the following sections.

5.1.1 Uncontrollable Model Dimensions. We found that the quantitative dimensions of the model were overlooked or difficult to reliably control through the design process. For D@a, the explicit dimension was not specified and it was not until *Physical Inspection* that we realized P@a was too small (12 mm in height). In contrast, D@b was constrained by multiple dimensions; however, P@b exhibited a smaller disk diameter (28 mm) and did not align with the intended diameter (40 mm).

These dimension-related issues would likely remain unnoticed for BLV users until the model is physically printed, as text-guided generation and chatbot-based descriptions do not reliably capture the model's quantitative dimensions. We expect these issues to be addressed by multimodal language models that can reason over dimensional information in 3D data, or by enabling prompt-based scaling at the slicing stage.

Reflection *Design processes should surface and communicate quantitative model attributes early by integrating dimension-aware feedback into generation and inspection stages.*

5.1.2 Unable to Edit Based on References. The inability to directly edit 3D models represents the primary constraint in iterative design workflows. Although our detour editing approach was technically feasible, we observed several quality limitations. Although the revised design RD@b visually matched our description, the 3D model regenerated from this image still exhibited sharp edges. This occurred because geometric filleting is difficult to capture through image-constrained generative processes alone.

Given that our study relied on existing open-source and commercial AI tools, text-guided image editing remains a practical strategy within current constraints. Despite the issues we described above, image-based editing allows for more targeted modifications while preserving other visual attributes compared to regenerating models solely through revised conversational prompts. We anticipate that as exemplary 3D model editing techniques (specifically within the realm of 3D diffusion models) become publicly available and usable, such limitations will be more readily addressed in the future.

Reflection *Iterative design processes should support reference-based modifications by enabling semantically guided editing that allows for precise and controllable adjustments.*

5.2 Inspection and Fabrication Observations

The printed artifacts generally met the functional and conceptual expectations of the designs and aligned with the impressions formed during conversational inspection. The key features of D@a (cat ears, weapon, and goggles) and D@b/RD@b (embossed Maine Coon face) that the chatbot described were confirmed to be clearly perceptible to touch. The printed artifacts aligned with our original design intentions as a game mascot and a keychain. We additionally found several preferences and issues in the functioning of 3D printing and printed products, as discussed in the following.

5.2.1 Rapid Tactile Prototyping for Inspection. The tactile silhouette patterns were implemented based on the RP@a and RP@b models for comparison, and they were printed significantly faster (each in less than 10 minutes) than the complete models (approximately 40 minutes and 28 minutes, respectively). These rapid inspection methods were effective in conveying coarse information about scale

and shape, but they were far less informative than a fully explorable physical model.

We think that similar inspection approaches offer stronger immediacy and expressive conveyance in real-time interfaces; however, in the context of rapid prototyping and the low material cost afforded by 3D printing, it will be more beneficial to rely directly on fully printed artifacts as the primary inspection medium. In such settings, the benefit of full 3D tactile information is likely to outweigh the additional time and material costs. We still consider the pre-print inspection to be valuable for long-duration or resource-intensive fabrication tasks. In such cases, alternative inspection kits, such as scaled-down models combined with silhouette representations, could support decision making about print tasks in the early stages.

An additional consideration in non-real-time inspection scenarios is how to determine shape-representative orientations for generating inspection references. This process may benefit from AI-assisted visual analysis that surveys the model to identify informative viewpoints for creating tactile patterns.

Reflection *Inspection strategies can directly leverage rapid full-scale printing to obtain informative tactile representations for small and quick tasks. For resource-intensive fabrication, downscaled or silhouette-based approximations should be considered as lightweight previews to support early-stage decision-making.*

5.2.2 Post-print Interpretation Challenges. We observed that uncertainty in interpreting printed results and performing post-print actions arose because the preceding design and inspection phases did not establish a sufficient 3D geometric mental representation for users.

We unintentionally printed P@a without support structures, leading to deformation of the overhang (**Figure 2 left**). This failure could not be identified because it was difficult to determine whether the observed printed structures were the result of fabrication errors or part of the original design. A related ambiguity arose during the removal of support structures, where the fabrication by-product and the intended features became indistinguishable by touch. RP@a was printed with tree support structures and their presence was communicated to all researchers; however, we initially struggled to distinguish them by touch alone. Through exploratory manipulation, we discovered that the support structures were relatively loose and that we were able to gently remove most of them by testing and wiggling. After this process, the tactile interpretation



Figure 2: Indistinguishable features observed in the study.

Left: the overhang deformation. Right: residual support structures that could not be removed.

aligned with our mental representation formed during the conversational inspection. Nevertheless, residual support structures remained in recessed and inaccessible regions for fingers that could not be removed despite repeated attempts, as shown in **Figure 2 right**.

In both of these cases, we observed a set of post-print ambiguities that manifest in tactile interpretation, including print failure deformation and residual fabrication by-products, both of which compromise the user's ability to interpret printed results. Together, these observations point to the lack of reliable reference information established during the design and inspection stages, which in turn led to challenges in interpreting printed results and performing post-print actions. This suggests the value of AI-assisted post-print inspection, which can compare printed artifacts with their original digital references and convey discrepancies in accessible forms. We further suggest encoding support structures in slicing scripts using distinguishable tactile patterns and possibly avoiding support generation in regions inaccessible to touch, allowing BLV users to independently identify and remove support material.

Reflection *Design and fabrication workflows should maintain a consistent reference understanding across stages by incorporating reference-aware post-print inspection mechanisms. For example, this can be achieved by using AI vision features to compare physical artifacts with their corresponding digital models.*

5.2.3 From Access to Personal Expression. Being exposed to hands-on personal fabrication experiences and supported by AI-assisted design tools also inspired additional imagination and interest in creating personally meaningful objects. For example, this experience led to our creative intentions regarding everyday household items such as bowls and cups, for personal use and sharing with friends. This suggests that establishing accessible entry points to personal fabrication, including appropriate tools and technical introductions, is essential to enable BLV users to engage in the ecosystem, explore diverse design processes, and express their needs within the system.

Reflection *Accessible fabrication systems should support sustained creative engagement by providing pathways that extend from initial access to ongoing personal expression.*

5.3 Conclusion and Future Work

As a milestone within a longer-term research agenda on supporting personal fabrication for BLV users, this work reports observations and design suggestions for potential AI-assisted accessibility features through our first-person technical designs and evaluations. Future work will focus on continually addressing the identified pain points through tool development and conducting formal studies involving more diverse communities to further investigate non-visual accessibility in personal fabrication.

6 Ethics Statement

This research follows a first-person methodology, focusing on the iterative analysis of the prototyped system through the authors' own design-making and reflective processes. The study was designed to establish the foundational implications of the design without the use of external participants. Consequently, no external user data was collected. We acknowledge that this limits our claims to the specific technical and physiological context of the authors within

the scope of this paper. As a starting point, this paper will inform a formal participatory study and technical evaluation with more diverse external stakeholders in future work.

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